

INFORMED SEARCH

Heuristics • Greedy Best-First • A* Search

Russell & Norvig | Bengaluru → Hyderabad classroom example

Using knowledge about the problem to guide the search

Learning Objectives

- Understand informed (heuristic) search.
- Define a heuristic function $h(n)$ and understand what it estimates.
- Explain Greedy Best-First Search and A^* Search.
- Understand the difference between $g(n)$, $h(n)$ and $f(n)$.
- Explain admissible and consistent heuristics.
- Trace Greedy and A^* on the Bengaluru $\rightarrow$ Hyderabad problem.

From Uninformed to Informed Search

The key difference is the use of additional knowledge.

UNINFORMED SEARCH

Uses only the problem definition.

Example: BFS, DFS, UCS.

INFORMED SEARCH

Uses a heuristic estimate to decide which states look promising.

Example: Greedy, A*.

Uninformed: “Which node should I explore?”

Informed: “Which node looks promising toward the goal?”

Heuristic = useful estimate about the remaining cost.

What Is a Heuristic Function?

DEFINITION

$h(n)$ estimates the cost of reaching the goal from node n .

IMPORTANT

It is an estimate — not necessarily the exact remaining cost.

$h(n)$ = estimated cost from $n \rightarrow$ goal

For Bengaluru $\rightarrow$ Hyderabad: $h(\text{Anantapur})$ could represent an estimate of the remaining travel cost.

A Common Route Problem

We use the same Bengaluru → Hyderabad problem.

START

Bengaluru

CURRENT

Anantapur

GOAL

Hyderabad

Two types of information

$g(n)$ = actual cost already travelled

$h(n)$ = estimated cost remaining to the goal

Informed search combines these pieces of information in different ways.

Three Numbers You Must Understand

$g(n)$

Actual path cost
from start $\rightarrow n$

$h(n)$

Estimated cost
from $n \rightarrow$ goal

$f(n)$

Evaluation value used
to choose nodes

A* uses: $f(n) = g(n) + h(n)$

This formula is the heart of A*.

Greedy Best-First Search

Choose the node that appears closest to the goal.

EVALUATION

$$f(n) = h(n)$$

QUESTION

“Which node looks closest to Hyderabad?”

- It ignores the path cost already travelled when choosing the next node.
- It can reach a goal quickly, but the solution is not guaranteed to be optimal.
- Uses a priority queue ordered by $h(n)$.

Greedy Example

Anantapur

$g(n) = 3$

$h(n) = 7$

Greedy value = 7

Chitradurga

$g(n) = 4$

$h(n) = 5$

Greedy value = 5

Decision

Compare $h(n)$:

7 vs 5

Choose Chitradurga

Greedy looks only at $h(n)$.

It does NOT ask how expensive the journey to the current node has already been.

A* Search

Combine the cost already travelled with the estimated remaining cost.

EVALUATION

$$f(n) = g(n) + h(n)$$

QUESTION

“What is my estimated total cost to the goal?”

$g(n)$ = actual cost so far **$h(n)$ = estimated cost remaining** **$f(n)$ = estimated total**

A* balances what has already happened + what is estimated to happen.

A* Example — Step by Step

Anantapur

$$g = 3$$

$$h = 7$$

$$f = 3 + 7 = 10$$

Chitradurga

$$g = 4$$

$$h = 5$$

$$f = 4 + 5 = 9$$

A* choice

Compare:

10 vs 9

Choose Chitradurga

A* chooses the smallest $f(n)$.

Notice: Greedy chose using h only; A* considers $g + h$.

Greedy Best-First vs A*

GREEDY BEST-FIRST

$$f(n) = h(n)$$

Looks only toward the goal.

Can be fast, but not guaranteed optimal.

A*

$$f(n) = g(n) + h(n)$$

Considers cost so far + estimated remaining cost.

Can be optimal under appropriate conditions.

Greedy = “looks closest” | A* = “best estimated total route”

Admissible Heuristic

A key condition for A* optimality.

DEFINITION

A heuristic is admissible if it never overestimates the true minimum remaining cost.

FORMULA

$$h(n) \leq h^*(n)$$

$h^*(n)$ = actual optimal cost from n to the goal.

Think: “The heuristic is optimistic.”

It may underestimate, but it should not overestimate the true cheapest remaining cost.

Why Admissibility Matters

SAFE ESTIMATE

If $h(n)$ never overestimates, A^* is not tricked into believing a costly path is better merely because its estimate is too small/large.

Example intuition: Actual remaining cost = 10

A^* RESULT

With the standard assumptions, an admissible heuristic supports optimal A^* tree search.

$h(n) = 7$ ✓ admissible

$h(n) = 13$ ✗ overestimate

Consistent Heuristic

Also called a monotonic heuristic.

FORMULA

$$h(n) \leq c(n, n') + h(n')$$

INTUITION

The estimated cost from n should not be greater than the cost of one step plus the estimate from the next node.

Consistency follows a triangle-inequality idea.

$n \rightarrow n' \rightarrow \text{goal}$ should not look cheaper than $n \rightarrow \text{goal}$ according to the heuristic.

Admissible vs Consistent

ADMISSIBLE

Compares $h(n)$ with the true optimal remaining cost:

$$h(n) \leq h^*(n)$$

“Never overestimate.”

CONSISTENT

Compares neighboring heuristic values:

$$h(n) \leq c(n, n') + h(n')$$

“Triangle inequality.”

For graph-search A*, consistency is a particularly useful condition because it supports nondecreasing f-values along paths.

Heuristic Examples

Good heuristics give useful guidance toward the goal.

STRAIGHT-LINE DISTANCE

For route finding, straight-line distance to the destination can be an intuitive lower-bound estimate.

MANHATTAN DISTANCE

For grid/puzzle movement, sum of horizontal + vertical distances is a common heuristic.

A good heuristic should be computationally inexpensive and informative.

More informed $\neq$ always “better” unless the required properties are preserved.

Informed Search — Complete Picture

HEURISTIC SEARCH

Uses $h(n)$ to guide search.

GREEDY

$$f(n) = h(n)$$

Goal-directed.

A*

$$f(n) = g(n) + h(n)$$

Balances actual + estimated cost.

A* + appropriate heuristic properties → optimality guarantees.

Admissible → never overestimate | Consistent → triangle inequality

Exam / Viva Focus

- What is a heuristic function $h(n)$?
- Differentiate $g(n)$, $h(n)$ and $f(n)$.
- Why is Greedy Best-First Search not guaranteed to be optimal?
- Write the A^* evaluation function.
- What is an admissible heuristic?
- What is a consistent heuristic?
- Differentiate admissible and consistent heuristics.
- Why is a good heuristic useful in search?

FINAL TAKEAWAY

Informed search uses an estimate to guide the search.

$h(n)$ → estimated remaining cost

Greedy → $h(n)$

A^* → $g(n) + h(n)$

Admissible → never overestimate

Consistent → triangle inequality

Next step: practice a complete Greedy and A^* search trace.